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Cable assembly with integral seal element

Abstract

An assembly is provided for an electric system. This electric system assembly includes an electric cable and a seal element. The seal element circumscribes and is configured integral with the electric cable. The seal element is configured from or otherwise includes a conductive polymer composite material.

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Background/Summary

BACKGROUND OF THE DISCLOSURE

1. Technical Field

(1) This disclosure relates generally to an electric system and, more particularly, to a cable assembly for the electric system.

2. Background Information

(2) An electric system such as a controller may include an internal compartment for housing an electric circuit. To communicate with device(s) outside of the internal compartment, a cable connected to the electric circuit may be passed through a hole in a wall of the housing. However, environmental debris (e.g., dust) and/or unwanted signals (e.g., noise) may also pass through the hole. Such debris and noise may interfere with proper operation of the electric circuit. It is known in the art therefore to provide the housing with a discrete gasket that engages the cable when the housing is assembled to seal, inter alia, the hole. Placement of such a gasket, however, may be difficult where the wall is located internally within the housing. It is also known to provide a sealed terminal (e.g., a receptacle) in the wall, where the cable within the compartment is mated to one side of the terminal and a cable outside of the compartment is mated to another side of the terminal. Such a sealed terminal, however, is often bulky and may not be suitable for electric systems with compact sizes. There is a need in the art therefore for an improved cable assembly.

SUMMARY OF THE DISCLOSURE

(3) According to an aspect of the present disclosure, an assembly is provided for an electric system. This electric system assembly includes an electric cable and a seal element. The seal element circumscribes and is configured integral with the electric cable. The seal element is configured from or otherwise includes a conductive polymer composite material.

(4) According to another aspect of the present disclosure, another assembly is provided for an electric system. This electric system assembly includes a housing and a cable assembly. The housing includes a first compartment, a second compartment and a wall. The wall is between and forms the first compartment and the second compartment. The wall is configured with an aperture. The cable assembly includes an electric cable and a seal element. The cable assembly is configured to prevent at least radio frequency signal transmission through the aperture. The electric cable extends through the aperture into the first compartment and the second compartment. The seal element is seated in the aperture and bonded to the electric cable.

(5) According to another aspect of the present disclosure, a manufacturing method is provided. During this manufacturing method, a seal element is formed integral with an electric cable to provide a cable assembly. The cable assembly is arranged with a housing. The housing includes a wall and an aperture extending through the wall. The electric cable projects through the aperture. The seal element is seated within the aperture such that the cable assembly plugs the aperture and is configured to prevent at least transmission of radio frequency signals through the aperture.

(6) The seal element may include a first segment and a second segment. The forming of the seal element may include bonding the electric cable to the first segment and the second segment, where the electric cable is between the first segment and the second segment.

(7) The seal element may be configured from or otherwise include a conductive polymer composite material.

(8) The electric cable may be configured as a flexible ribbon cable.

(9) The electric cable may include a sheath and an electric conductor within the sheath. The seal element may be bonded directly to the sheath.

(10) The conductive polymer composite material may include a polymer foam and a plurality of metal particles embedded within the polymer foam.

(11) The conductive polymer composite material may be configured to block transmission of radio frequency (RF) signals.

(12) The seal element may include a first segment and a second segment. The electric cable may be

between the first segment and the second segment.

(13) The seal element may have a polygonal cross-sectional geometry.

(14) The electric system assembly may include a conductive adhesive. This conductive adhesive may bond the seal element to the electric cable.

(15) The electric system assembly may include a housing. This housing may include a wall. The electric cable may pass through an aperture in the wall. The seal element may be seated in the aperture. The seal element may be configured to plug an annular gap between the electric cable and the housing.

(16) The wall may have a first face and a second face. The aperture may extend through the wall between the first face and the second face. A first portion of the seal element may project out from the first face. In addition or alternatively, a second portion of the seal element may project out from the second face.

(17) The aperture may be configured as or otherwise include a dovetailed slot in the wall.

(18) The housing may also include a lid. The seal element may be sandwiched between the wall and the lid.

(19) The housing may be configured with a first compartment and a second compartment. The wall may be between the first compartment and the second compartment. A first end of the electric cable may be within the first compartment. In addition or alternatively, a second end of the electric cable may be within the second compartment.

(20) The seal element may circumscribe the electric cable.

(21) The seal element may be configured from or otherwise include a conductive polymer composite material.

(22) The seal element may include a first segment and a second segment. The first segment and the second segment may be seated within the aperture. The electric cable may be between the first segment and the second segment.

(23) The present disclosure may include any one or more of the individual features disclosed above and/or below alone or in any combination thereof.

(24) The foregoing features and the operation of the invention will become more apparent in light of the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic illustration of an electric system.

(2) FIG. 2 is a sectional illustration of a housing for the electric system in an x-z plane.

(3) FIG. 3 is a sectional illustration of the electric system housing in an x-y plane taken along line 3-3 in FIG. 2.

(4) FIG. 4 is a sectional illustration of the electric system housing in a y-z plane taken along line 4-4 in FIG. 2.

(5) FIG. 5 is a perspective illustration of the electric system shown without a lid on a base of the electric system housing.

(6) FIG. 6 is an illustration of a cable assembly with hidden internal electric conductors schematically shown by dashed lines.

(7) FIG. 7A is cross-sectional illustration of an electric cable for the cable assembly taken along line 7A-7A in FIG. 6.

(8) FIG. 7B is a cross-sectional illustration of another electric cable for the cable assembly taken along line 7B-7B in FIG. 6.

(9) FIG. 8 is a cross-sectional illustration of the cable assembly taken along line 8-8 in FIG. 6, where the electric cable is shown without its internal parts for ease of illustration.

- (10) FIG. **9** is a sectional illustration of a portion of seal element material.
- (11) FIG. **10** is a partial sectional illustration of the electric system in the y-z plane.
- (12) FIG. **11** is a partial sectional illustration of the electric system in the x-z plane.
- (13) FIG. **12** schematically illustrates a perimeter of a slot/aperture and a perimeter of a seal element.
- (14) FIG. **13** is a partial sectional illustration of the electric system housing in the y-z plane.
- (15) FIG. **14** is a cross-sectional illustration of the cable assembly.
- (16) FIG. **15** is a flow diagram of a method for manufacturing the electric system.

DETAILED DESCRIPTION

- (17) FIG. **1** illustrates an electric system **20**. This electric system **20** may be configured as or otherwise include an electrical system and/or an electronic system. The electric system **20**, for example, may be configured as a controller for an aerial, ground and/or aquatic vehicle or device. The present disclosure, however, is not limited to such an exemplary electric system configuration. The electric system **20**, for example, may also or alternatively be configured as or otherwise include a communication system, a monitoring system, a junction box and/or any other electric system that may utilize one or more cable assemblies as disclosed herein.
- (18) The electric system **20** of FIG. **1** includes a first electric component **22**, a second electric component **24** and one or more unitary, standalone cable assemblies **26A** and **26B** (generally referred to as “**26**”). The electric system **20** of FIG. **1** also includes an electric system housing **28** for housing one or more of the electric system parts **22**, **24** and **26**.
- (19) Referring to FIG. **2**, the electric system housing **28** is configured with one or more internal compartments **30A** and **30B** (generally referred to as “**30**”); e.g., chambers, cavities, etc. The internal compartments **30** of FIG. **2** are collectively formed by a base **32** and a lid **34** of the electrical system housing **28**.
- (20) The housing base **32** of FIG. **2** includes a base wall **36** (e.g., a bottom wall), an outer perimeter wall **38** (e.g., an exterior wall) and at least one inner wall **40** (e.g., an interior wall, a bulkhead, a divider, etc.). The base wall **36** may be a planar wall located at a first (e.g., bottom) end **42** of the electric system housing **28**. Referring to FIG. **3**, the base wall **36** extends lengthwise (e.g., along an x-axis) between opposing first and second sides **44** and **46** of the electric system housing **28**. The base wall **36** extends widthwise (e.g., along a y-axis) between opposing third and fourth sides **48** and **50** of the electric system housing **28**.
- (21) The perimeter wall **38** is connected to base wall **36**. The perimeter wall **38** is located at (e.g., on, adjacent or proximate) one or more of the housing sides **44**, **46**, **48** and **50**. The perimeter wall **38** has a (e.g., rectangular) tubular configuration to provide a peripheral boundary for the electric system housing **28** in a lengthwise-widthwise (e.g., x-y) plane. The perimeter wall **38** of FIG. **3**, for example, includes a plurality of perimeter wall segments **52-55**. The first wall segment **52** is located at the housing first side **44**. The second wall segment **53** is located at the housing second side **46**. The third wall segment **54** is located at the housing third side **48**. The fourth wall segment **55** is located at the housing fourth side **50**. Each of the first wall segment **52** and the second wall segment **53** of FIG. **3** extends widthwise (e.g., along the y-axis) between and is connected to the third wall segment **54** and the fourth wall segment **55**. Each of the third wall segment **54** and the fourth wall segment **55** of FIG. **3** extends lengthwise (e.g., along the x-axis) between and is connected to the first wall segment **52** and the second wall segment **53**. Referring to FIG. **2**, each of the wall segments **52-55** projects vertically (e.g., along a z-axis) out from the base wall **36** to a distal end **58** of the perimeter wall **38**, which perimeter wall distal end **58** may be proximate a second (e.g., top) end **60** of the electric system housing **28**.
- (22) The inner wall **40** is connected to the base wall **36**. The inner wall **40** projects vertically (e.g., along the z-axis) out from the base wall **36** to a distal end **62** of the inner wall **40**. This inner wall distal end **62** may be co-planar with the perimeter wall distal end **58** (see also FIGS. **3** and **4**). The inner wall **40** has and extends lengthwise (e.g., along the x-axis) between opposing faces **64** and

66; e.g., wall surfaces. Referring to FIG. 3, the inner wall **40** extends widthwise (e.g., along the y-axis) between and is connected to the third wall segment **54** and the fourth wall segment **55**; see also FIG. 4.

(23) The inner wall **40** of FIG. 3 includes one or more slots **68A** and **68B** (generally referred to as “**68**”); e.g., channels, notches, etc. These slots **68** are located at respective discrete widthwise (e.g., y-axis) locations along the inner wall **40** between the third wall segment **54** and the fourth wall segment **55**. Each of these slots **68** extends lengthwise (e.g., along the x-axis) through the inner wall **40** between the inner wall first face **64** and the inner wall second face **66**. Each of the slots **68** extends widthwise (e.g., along the y-axis) within the inner wall **40** between respective opposing slot side surfaces **70A**, **70B** (generally referred to as “**70**”) and **72A**, **72B** (generally referred to as “**72**”); see also FIG. 4. Referring to FIG. 4, each of the slots **68** extends vertically (e.g., along the z-axis) into the inner wall **40** from the inner wall distal end **62** to a respective slot end surface **74A**, **74B** (generally referred to as “**74**”).

(24) Referring to FIGS. 2 and 4, the housing lid **34** is configured to enclose an open end of the housing base **32** at the housing second end **60**. The housing lid **34** of FIGS. 2 and 4, for example, is configured as a planar member that engages (e.g., vertically seals against, contacts, etc.) the housing base **32** and, more particularly, one or more or all of the housing base parts **40** and **52-55**. The housing lid **34** may be removably attached (e.g., mechanically fastened) to the housing base **32**. The housing lid **34** of FIG. 2 extends lengthwise (e.g., along the x-axis) between and to the housing first side **44** and the housing second side **46**. The housing lid **34** of FIG. 4 extends widthwise (e.g., along the y-axis) between and to the housing third side **48** and the housing fourth side **50**.

(25) Referring to FIG. 3, the first compartment **30A** extends lengthwise (e.g., along the x-axis) between and is formed by first wall segment **52** and the inner wall **40** and its first face **64**. The first compartment **30A** extends widthwise (e.g., along the y-axis) between and is formed by the third wall segment **54** and the fourth wall segment **55**. Referring to FIG. 2, the first compartment **30A** extends vertically (e.g., along the z-axis) between and is formed by the base wall **36** and the housing lid **34**.

(26) Referring to FIG. 3, the second compartment **30B** extends lengthwise (e.g., along the x-axis) between and is formed by second wall segment **53** and the inner wall **40** and its second face **66**. The second compartment **30B** extends widthwise (e.g., along the y-axis) between and is formed by the third wall segment **54** and the fourth wall segment **55**. Referring to FIG. 2, the second compartment **30B** extends vertically (e.g., along the z-axis) between and is formed by the base wall **36** and the housing lid **34**.

(27) Referring to FIG. 4, the first slot **68A** is enclosed by the housing lid **34**. The housing lid **34** and the inner wall **40** may thereby collectively form a first aperture **76A** (e.g., a through-hole) within the electric system housing **28**. Before the first slot **68A** receives the first cable assembly **26A** as described below (see FIG. 5), the first aperture **76A**/the first slot **68A** fluidly couple the first compartment **30A** and the second compartment **30B** together; see also FIG. 3.

(28) The second slot **68B** is enclosed by the housing lid **34**. The housing lid **34** and the inner wall **40** may thereby collectively form a second aperture **76B** (e.g., a through-hole) within the electric system housing **28**. Before the second slot **68B** receives the second cable assembly **26B** as described below (see FIG. 5), the second aperture **76B**/the second slot **68B** fluidly couple the first compartment **30A** and the second compartment **30B** together; see also FIG. 3.

(29) The electric system housing **28** and each of its parts **32** and **34** may be constructed from a conductive material such as, but not limited to, metal. Examples of the conductive material may include, but are not limited to, aluminum (Al), copper (Cu), nickel (Ni), titanium (Ti), carbon (C) and a (e.g., corrosion resistant) metal alloy such as an alloy including one or more of the foregoing metals. The conductive material may be coated with a protective coating, which coating may be selected to provide the respective component (e.g., **28**, **32**, **34**) with corrosion protection.

(30) Referring to FIG. 5, the first electric component **22** may be configured as or otherwise include an electronic circuit. The first electric component **22** of FIG. 5, for example, includes a printed circuit board (PCB) **78** and one or more circuit elements (e.g., **80A-H**; generally referred to as “**80**”); e.g., electronic components. These circuit elements **80** are mounted to the printed circuit board **78**. Each of the circuit elements **80** is electrically coupled with one or more respective conductive paths (e.g., tracks) on the printed circuit board **78**. Examples of the circuit elements **80** include, but are not limited to, processing device(s), memory device(s), resistor(s), capacitor(s), transistor(s), inductor(s), diode(s), switch(es), input device(s), output device(s), etc. Each circuit element **80A** and **80B** of FIG. 2, in particular, is configured as a terminal **82A**, **82B** (generally referred to as “**82**”); e.g., an input and/or output (I/O) device such as, but not limited to, a receptacle.

(31) The first electric component **22** is arranged within the first compartment **30A**. The first electric component **22** is mounted to the electric system housing **28** and, more particularly, the base wall **36** for example.

(32) The second electric component **24** may be configured as or otherwise include an electronic circuit. The second electric component **24** of FIG. 5, for example, includes a printed circuit board (PCB) **84** and one or more circuit elements (e.g., **86A-F**; generally referred to as “**86**”); e.g., electronic components. These circuit elements **86** are mounted to the printed circuit board **84**. Each of the circuit elements **86** is electrically coupled with one or more respective conductive paths (e.g., tracks) on the printed circuit board **84**. Examples of the circuit elements **86** include, but are not limited to, processing device(s), memory device(s), resistor(s), capacitor(s), transistor(s), inductor(s), diode(s), switch(es), input device(s), output device(s), etc. Each circuit element **86** of FIG. 2, in particular, is configured as a terminal **88A**, **88B** (generally referred to as “**88**”); e.g., an input and/or output (I/O) device such as, but not limited to, a receptacle.

(33) The second electric component **24** is arranged within the second compartment **30B**. The second electric component **24** is mounted to the electric system housing **28** and, more particularly, the base wall **36** for example.

(34) Each of the cable assemblies **26A**, **26B** includes an (e.g., flexible) electric cable **90A**, **90B** (generally referred to as “**90**”) and a seal element **92A**, **92B** (generally referred to as “**92**”). Each of the cable assemblies **26** of FIG. 5 also includes one or more terminals **94A**, **94B** (generally referred to as “**94**”) and **96A**, **96B** (generally referred to as “**96**”); e.g., input and/or output (I/O) devices such as, but not limited to, plugs.

(35) Referring to FIG. 6, the electric cable **90** is adapted for transferring electricity and/or communicating signals. This electric cable **90** may be a (e.g., flexible) ribbon cable such as, but not limited to, a (e.g., shielded) coaxial ribbon or a (e.g., shielded) flex cable. The electric cable **90** of FIG. 6 has a length that extends longitudinally along a longitudinal centerline **98** of the respective cable assembly **26** between opposing longitudinal ends **100** and **102** of the electric cable **90**.

(36) The electric cable **90** of FIG. 6 includes a sheath **104** and one or more electric conductors **106** at least partially disposed within and/or covered by the sheath **104**. The electric conductors **106** may be configured as conductive wires or conductive layers; e.g., laminae within a laminate. Each electric conductor **106** may be constructed from electrically conductive material such as metal; e.g., copper (Cu), etc. The sheath **104** may be constructed from dielectric/electrically non-conductive material such as polymer.

(37) Referring to FIGS. 7A and 7B, the electric cable **90** may also include shielding material **108** such as, but not limited to, woven or non-woven metal. This shielding material **108** is adapted to shield one or more of the electric conductors **106** from outside interference. Referring to FIG. 7A, the shielding material **108** may be wrapped around each respective electric conductor **106**.

Referring to FIG. 7B, the shielding material **108** may alternatively (or also) be wrapped around a grouping of the electric conductors **106**. The shielding material **108** of FIGS. 7A and 7B may be embedded within or otherwise covered by the sheath **104**.

(38) Referring to FIG. 6, the seal element **92** is located longitudinally along the centerline **98** intermediately between the cable first end **100** and the cable second end **102**; e.g., approximately midway between the ends **100** and **102**. The seal element **92** is configured integral with the electric cable **90**. The seal element **92**, for example, may be bonded (e.g., adhered) to the electric cable **90**. Alternatively, the seal element **92** may be co-molded onto the electric cable **90**.

(39) The seal element **92** of FIG. 8 wraps around/circumscribes an intermediate portion of the electric cable **90**. The seal element **92** extends laterally between and to opposing first and second sides **110** and **112** of the seal element **92**. The seal element **92** extends transversely between and to opposing third and fourth sides **114** and **116** of the seal element **92**, where each of the seal element sides **114** and **116** extend laterally between and to the seal element sides **110** and **112**. Referring to FIG. 6, the seal element **92** extends longitudinally along the centerline **98** between and to opposing first and second end **118** and **120** of the seal element **92**, where each of the seal element sides **110**, **112**, **114** and **116** extends longitudinally between and to the seal element ends **118** and **120**.

(40) The seal element **92** of FIG. 8 include a plurality of segments **122** and **124**; e.g., halves. The seal element first segment **122** is located at the seal element third side **114**. The seal element second segment **124** is located at the seal element fourth side **116**. Each seal element segment **122**, **124** extends laterally between and to the seal element sides **110** and **112**. Thus, side surfaces **126A** and **126B**, **128A** and **128B** of the seal element segments **122** and **124** are respectively flush at the seal element sides **110**, **112**. Each seal element segment **122**, **124** extends transversely from a respective seal element side **114**, **116** to a respective interface **130**, **132**. Referring to FIG. 6, each seal element segment **122**, **124** (see **124** in FIG. 8) extends longitudinally between and to the seal element ends **118** and **120**.

(41) Referring to FIG. 8, the seal element first segment **122** is arranged next to a first side **134** of the electric cable **90**. The first segment interface **130** is bonded (e.g., directly) to the electric cable **90**/the sheath **104** by, for example, an electrically conductive adhesive; e.g., a conductive non-corrosive and/or pressure sensitive adhesive. Examples of the electrically conductive adhesive may include, but are not limited to, an electrically conductive silicon adhesive and an electrically conductive room-temperature-vulcanized (RTV) adhesive; e.g., electrically conductive RTV silicon adhesive. The seal element second segment **124** is arranged next to a second side **136** of the electric cable **90** as well as the seal element first segment **122** and its interface **130**. The second segment interface **132** is bonded (e.g., directly) to the electric cable **90**/the sheath **104** and the first segment interface **130** by, for example, the conductive adhesive. The intermediate portion of the electric cable **90** is disposed (e.g., sandwiched) between the seal segment segments **122** and **124**. The intermediate portion of the electric cable **90** is thereby disposed laterally and transversely within, and longitudinally covered by, the seal element **92**.

(42) The seal element **92** and its segments **122** and **124** may be constructed from material adapted to block transmission therethrough of signal noise; e.g., electromagnetic interference (EMI) such as, but not limited to, at least radio frequency (RF) signals. The seal element **92** and its segment **122** and **124**, for example, may be constructed from a conductive polymer composite material. Referring to FIG. 9, this composite material may include a (e.g., closed cell) polymer foam **138** and a plurality of conductive (e.g., metal) particles **140** embedded within the polymer foam **138**. An example of the polymer foam **138** is a (e.g., conductive, low density) urethane foam. An example of the conductive particle **140** is a silver (Ag) plated, copper (Cu) particle. The present disclosure, however, is not limited to the foregoing exemplary foam/particle materials. Furthermore, the seal element material may also be adapted to block passage therethrough of certain environmental elements; e.g., dust, moisture, etc.

(43) Referring to FIG. 10, each of the cable assemblies **26** is mated with a respective one of the slots **68**/apertures **76A** and **76B** (generally referred to as “**76**”). In particular, each seal element **92** and its segments **122** and **124** is seated within a respective one of the slots **68**/apertures **76**. Referring to FIGS. 10 and 11, each seal element third side **114** engages (e.g., is pressed vertically

against) a respective one of the slot end surface **74**. Each seal element fourth side **116** engages (e.g., is pressed vertically against) and interior surface **142** of the housing lid **34**. Referring to FIG. **10**, each seal element side **110**, **112** engages (e.g., is pressed widthwise against) a respective one of the slot side surfaces **70**, **72**. Each seal element **92** thereby seals/plugs a (e.g., annular) gap between a respective one of the electric cables **90** and the electric system housing **28**. In other words, each cable assembly **26** (e.g., a combination of the seal element **92** and the electric cable **90**) seals/plugs a respective one of the apertures **76**. With this arrangement, referring to FIG. **11**, each electric cable **90** passes (e.g., extends, projects) through a respective one of the apertures **76** from the first compartment **30A** to the second compartment **30B**, while the respective cable assembly **26** and its seal element **92** prevent or reduce transmission of unwanted signals (e.g., RF signals) between the compartments **30** through the respective aperture **76**.

(44) A length **144** of each seal element **92** may be greater than a width **146** of the inner wall **40**. A first portion of each seal element **92** of FIG. **11** projects out from the inner wall first face **64** into the first compartment **30A** to the end **118**. A second portion of each seal element **92** of FIG. **11** also (or alternatively) projects out from the inner wall second face **66** into the second compartment **30B** to the end **120**.

(45) Referring to FIG. **12**, dimensions (e.g., width **148** and/or height **150**) of each seal element **92** may be sized greater than corresponding dimensions (e.g., width **152** and/or height **154**) of each respective aperture **76** before assembly of the respective cable assembly **26** (see FIG. **6**) with the electric system housing **28** (see FIGS. **2-4**). Thus, when each seal element **92** is inserted into/seated within the respective slot **68** of FIGS. **10** and **11**, the seal element material is (e.g., slightly) compressed and pressed against the housing parts **32** and **34**.

(46) Referring again to FIG. **5**, a first portion and end of each electric cable **90** is disposed within the first compartment **30A**. A second portion and end of each electric cable **90** is disposed within the second compartment **30B**. Each of the terminals **94**, **96** is mated with (e.g., plugged into) a respective one of the terminals **82**, **88**. Each cable assembly **26** thereby electrically couples the first electric component **22** in the first compartment **30A** to the second electric component **24** in the second compartment **30B**.

(47) In some embodiments, referring to FIG. **4**, each slot **68** and associated aperture **76** may have a polygonal cross-sectional geometry when viewed, for example, in a plane perpendicular to the length of the associated electric cable **90** and/or the longitudinal centerline **98** of the associated cable assembly **26** (see FIG. **6**); e.g., plane of FIG. **4**. This polygonal cross-sectional geometry may be a rectangular cross-sectional geometry as shown in FIG. **4**. Alternatively, referring to FIG. **13**, the polygonal cross-sectional geometry may be a (e.g., isosceles) trapezoidal cross-sectional geometry. Each slot **68**, for example, may be configured as a dovetail slot. The present disclosure, of course, is not limited to the foregoing exemplary cross-sectional geometries.

(48) In some embodiments, referring to FIG. **8**, each seal element **92** may have a polygonal cross-sectional geometry when viewed, for example, in a plane perpendicular to the length of the associated electric cable **90** and/or the longitudinal centerline **98** of the associated cable assembly **26**; e.g., plane of FIG. **8**. This polygonal cross-sectional geometry may be a rectangular cross-sectional geometry as shown in FIG. **8**. Alternatively, referring to FIG. **14**, the polygonal cross-sectional geometry may be a (e.g., isosceles) trapezoidal cross-sectional geometry. The present disclosure, of course, is not limited to the foregoing exemplary cross-sectional geometries.

(49) In some embodiments, each seal element **92** and each associated slot **68**/aperture **76** may have similar/common cross-sectional geometry shapes. For example, each seal element **92** may have the cross-sectional geometry shape of FIG. **8** or FIG. **14** and each associated slot **68**/aperture **76** may have the cross-sectional geometry shape of FIG. **4** or FIG. **13**. However, in other embodiments, at least one of the seal elements **92** may have a different cross-sectional geometry shape than the associated slot **68**/aperture **76**. For example, each seal element **92** may have the cross-sectional geometry shape of FIG. **8** and each associated slot **68**/aperture **76** may have the cross-sectional

geometry shape of FIG. 13. With such an arrangement, the side surfaces **70** and **72** of each slot **68** may retain the associated cable assembly **26** during assembly (e.g., after mating with the slot **68** but before placement of the housing lid **34**).

(50) FIG. 15 is a flow diagram of a method **1500** for manufacturing an electric system. For ease of description, the electric system is described below as the electric system **20** described above. However, the manufacturing method **1500** is not limited to manufacturing such an exemplary electric system.

(51) In step **1502**, the electric system housing **28** is provided.

(52) In step **1504**, the first electric component **22** is provided.

(53) In step **1506**, the first electric component **22** is arranged within the first compartment **30A**.

(54) In step **1508**, the second electric component **24** is provided.

(55) In step **1510**, the second electric component **24** is arranged within the second compartment **30B**.

(56) In step **1512**, the first electric cable **90A** is provided.

(57) In step **1514**, the first seal element **92A** is formed integral with the first electric cable **90A**. The conductive adhesive, for example, may be applied to the first segment interface **130** of the seal element first segment **122**. The first electric cable **90A** may be arranged next to the seal element first segment **122** and bonded to the first segment interface **130** by the conductive adhesive. The conductive adhesive may also be applied to the interface **132** of the seal element second segment **124**. This seal element second segment **124** may be arranged next to the first electric cable **90A** and the seal element first segment **122**. The conductive adhesive may bond the seal element second segment **124** to the first electric cable **90A** as well as bond and electrically couple the seal element second segment **124** to the seal element first segment **122**. Subsequently, the bonded seal element segments **122** and **124** may be cut (e.g., die cut) to provide the flush surfaces **126A** and **126B**, **128A** and **128B** at the first seal element sides **110**, **112**.

(58) The bonding of the first seal element **92A** to the first electric cable **90A** may (e.g., permanently) affix the first seal element **92A** to the first electric cable **90A**. The bonding thereby forms the first seal element **92A** and the first electric cable **90A** into the unitary, standalone first cable assembly **26A**. In other words, the first cable assembly **26A** may be a complete unit, and may be moved as a standalone body. This is in contrast to, for example, a discrete seal element that is only arranged with a cable once the cable is arranged with a housing. This is also in contrast to, for example, a discrete seal element that is also used to seal a rim of a housing to a lid.

(59) In step **1516**, the second electric cable **90B** is provided.

(60) In step **1518**, the second seal element **92B** is formed integral with the second electric cable **90B**. The conductive adhesive, for example, may be applied to the interface **130** of the seal element first segment **122**. The second electric cable **90B** may be arranged next to the seal element first segment **122** and bonded to the first segment interface **130** by the conductive adhesive. The conductive adhesive may also be applied to the interface **132** of the seal element second segment **124**. This seal element second segment **124** may be arranged next to the second electric cable **90B** and the seal element first segment **122**. The conductive adhesive may bond the seal element second segment **124** to the second electric cable **90B** as well as bond and electrically couple the seal element second segment **124** to the seal element first segment **122**. Subsequently, the bonded seal element segments **122** and **124** may be cut (e.g., die cut) to provide the flush surfaces **126A** and **126B**, **128A** and **128B** at the second seal element sides **110**, **112**.

(61) The bonding of the second seal element **92B** to the second electric cable **90B** may (e.g., permanently) affix the second seal element **92B** to the second electric cable **90B**. The bonding thereby forms the second seal element **92B** and the second electric cable **90B** into the unitary, standalone second cable assembly **26B**. In other words, the second cable assembly **26B** may be a complete unit, and may be moved as a standalone body. This is in contrast to, for example, a discrete seal element that is only arranged with a cable once the cable is arranged with a housing.

This is also in contrast to, for example, a discrete seal element that is also used to seal a rim of a housing to a lid.

(62) In step **1520**, each cable assembly **26** is electrically coupled to the first electric component **22** and the second electric component **24**, for example, as described above.

(63) In step **1522**, the first cable assembly **26A** and the second cable assembly **26B** are arranged with the electric system housing **28**. The first seal element **92A**, for example, may be seated within the first slot **68A** as described above. Similarly, the second seal element **92B** may be seated within the second slot **68B** as described above. The housing lid **34** may subsequently be attached to the housing base **32**, which thereby sandwiches each seal element **92** between the housing lid **34** and the housing base **32** and its inner wall **40**.

(64) While various embodiments of the present disclosure have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the disclosure. For example, the present disclosure as described herein includes several aspects and embodiments that include particular features. Although these features may be described individually, it is within the scope of the present disclosure that some or all of these features may be combined with any one of the aspects and remain within the scope of the disclosure. Accordingly, the present disclosure is not to be restricted except in light of the attached claims and their equivalents.

Claims

1. An assembly for an electric system, comprising: a first electric component; a second electric component; a housing configured with a first compartment and a second compartment, the housing including a wall between the first compartment and the second compartment, the first electric component arranged within the first compartment, and the second electric component arranged within the second compartment; a cable assembly comprising one or more terminals and an electric cable passing through an aperture in the wall, the electric cable extending lengthwise from a first end to a second end, wherein the second end opposes the first end, the first end of the electric cable within the first compartment, and the second end of the electric cable within the second compartment; a seal element circumscribing and configured integral with the electric cable, the seal element comprising a conductive polymer composite material, the seal element seated in the aperture, and the seal element configured to plug an annular gap between the electric cable and the housing, and a conductive adhesive bonding the seal element to the electric cable, wherein at least one of the first electric component or the second electric component comprises a printed circuit board.
2. The assembly of claim 1, wherein the electric cable is configured as a flexible ribbon cable.
3. The assembly of claim 1, wherein the conductive polymer composite material is configured to block transmission of radio frequency (RF) signals.
4. The assembly of claim 1, wherein the seal element includes a first segment and a second segment; and the electric cable is between the first segment and the second segment.
5. The assembly of claim 1, wherein the seal element has a polygonal cross-sectional geometry.
6. The assembly of claim 1, wherein the seal element has a seal element length; the wall has a wall width; and the seal element length is greater than the wall width.
7. The assembly of claim 1, wherein the seal element has a seal element width in a non-assembled configuration; the aperture has an aperture width in the non-assembled configuration; and the seal element width is greater than the aperture width.
8. The assembly of claim 7, wherein the seal element is compressed when positioned within the aperture in an assembled configuration.
9. The assembly of claim 1, wherein the seal element has a seal element cross sectional geometry with a first shape; and the aperture has an aperture cross sectional geometry with a second shape

that is different from the first shape.

10. The assembly of claim 1, wherein the first electric component includes a first printed circuit board, and wherein the second electric component includes a second printed circuit board.

11. An assembly for an electric system, comprising: a first electric component; a second electric component; a housing configured with a first compartment and a second compartment, the first electric component arranged within the first compartment, and the second electric component arranged within the second compartment; a cable assembly comprising one or more terminals and an electric cable, the electric cable comprising a sheath, one or more electric conductors at least partially disposed within and/or covered by the sheath, and shielding material adapted to shield the one or more electric conductors from outside interference; and a seal element circumscribing and configured integral with the electric cable, the seal element comprising a conductive polymer composite material; the seal element adhered or co-molded directly onto and covering a portion of the electric cable, wherein: the seal element comprises a first segment and a second segment, and a first segment interface is directly bonded to the electric cable by an electrically conductive adhesive, and a second segment interface is directly bonded to the electric cable and to the first segment interface by the electrically conductive adhesive such that a portion of the electric cable is disposed laterally and transversely within, and is longitudinally covered by, the seal element; and at least one of the first electric component or the second electric component comprises a printed circuit board.

12. The assembly of claim 11, wherein the housing includes a wall between the first compartment and the second compartment; the electric cable passing through an aperture in the wall; and the seal element seated in the aperture, and the seal element configured to plug an annular gap between the electric cable and the housing.

13. The assembly of claim 12, wherein the wall has a first face and a second face, and the aperture extends through the wall between the first face and the second face; and at least one of the first segment of the seal element projects out from the first face; or the second segment of the seal element projects out from the second face.

14. The assembly of claim 12, wherein the aperture comprises a dovetailed slot in the wall.

15. The assembly of claim 12, wherein the housing further includes a lid; and the seal element is sandwiched between the wall and the lid.

16. The assembly of claim 12, wherein the electric cable extends lengthwise from a first end to a second end, wherein the second end opposes the first end; the first end of the electric cable is within the first compartment; and the second end of the electric cable is within the second compartment.

17. The assembly of claim 11, wherein the seal element is adhered directly onto the sheath with a conductive adhesive.

18. An assembly for an electric system, comprising: a first electric component; a second electric component; a housing configured with a first compartment and a second compartment, the first electric component arranged within the first compartment, and the second electric component arranged within the second compartment; a cable assembly comprising one or more terminals and an electric cable, the electric cable comprising a sheath, one or more electric conductors at least partially disposed within and/or covered by the sheath, and shielding material adapted to shield the one or more electric conductors from outside interference; and a seal element circumscribing and configured integral with the electric cable, the seal element comprising a conductive polymer composite material; the conductive polymer composite material comprising a polymer foam and a plurality of metal particles embedded within the polymer foam, wherein the seal element comprises a first segment and a second segment, a first segment interface directly bonded to the electric cable by an electrically conductive adhesive, and a second segment interface directly bonded to the electric cable and to the first segment interface by the electrically conductive adhesive, such that a portion of the electric cable is disposed laterally and transversely within, and is longitudinally

covered by, the seal element; and wherein at least one of the first electric component or the second electric component comprises a printed circuit board.
